F1 Race Outcome Prediction Using Machine Learning

1. Data Collection & Understanding

1.1 Data Source

The dataset consists of official Formula 1 lap-level race data collected from seasons **2018 to 2025 (inclusive)**.
Each season file contains detailed lap-by-lap telemetry and race results.
The dataset is relevant and reliable because:

- It contains official race timing and classification data.
- It spans multiple regulation eras (hybrid era dominance and post-2022 regulation reset).
- It includes performance, environmental, and result-based attributes.

1.2 Dataset Description

Two primary datasets were constructed:
1. Lap-Level Dataset (`f1_cleaned_laps.csv`)

- 175,763 records
- Granularity: Driver × Lap
- Key Attributes:
  - grid_position
  - lap_time
  - sector times
  - tyre data
  - pit information
  - air & track temperature
  - rainfall
  - finish_position (repeated per lap)

  2. Aggregated Driver-Race Dataset (`f1_driver_race.csv`)
- 2,853 records
- Granularity: Driver × Race
- Key Features:
  - grid_position
  - finish_position (target)
  - avg_lap_time, std_lap_time, best_lap_time
  - pit_stop_count
  - position_gain
  - avg_air_temp, avg_track_temp
  - rain_probability

  This aggregated dataset was used for modeling.

1.3 Ethics & Bias Considerations

This dataset does not contain personal or sensitive information. However, structural biases exist:

- **Team Dominance Bias**: Certain constructors (e.g., dominant eras) heavily influence outcomes.
- **Regulation Changes**: Performance patterns shift between regulation eras (2018–2021 vs 2022+).
- **Survivorship Bias**: DNFs and mechanical failures introduce unpredictability not fully modeled.
  The model predicts statistical tendencies, not deterministic outcomes.

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2. Problem Understanding & Objective Definition

2.1 Problem Statement

The objective is to predict a driver's **final finishing position** in a Formula 1 race using pre-race and historical performance data.
This is framed as a **regression problem**, where:
Target Variable: `finish_position`

2.2 Objectives

- Build a pipeline from lap-level raw data to race-level features.
- Engineer meaningful historical performance metrics.
- Train and evaluate predictive models using time-aware validation.
- Identify the most influential factors in race outcomes.

2.3 Assumptions

- Grid position is known before race start.
- Historical driver and team averages reflect skill and machinery strength.
- Weather variables are known before or during the race weekend.
- The model predicts finishing order, not real-time strategy changes.

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3. Data Preparation & Preprocessing

3.1 Data Cleaning

- Converted lap-level pit times into a binary `pitted` feature.
- Converted rainfall to numeric probability.
- Converted event_date to datetime and sorted chronologically.
- Aggregated lap-level data into race-level metrics.
  Missing values:
- Cumulative statistics were shifted to prevent data leakage.
- Initial NaNs for first races were filled with 0 (cold-start condition).

3.2 Aggregation Strategy

Lap-level features were aggregated as:

Feature	Aggregation
avg_lap_time	mean
std_lap_time	standard deviation
best_lap_time	minimum
pit_stop_count	sum
finish_position	last

3.3 Feature Engineering

To quantify driver and team performance:

- `past_avg_pos` → Rolling mean of previous finish positions (Driver Skill)
- `past_avg_points` → Rolling mean of points scored
- `team_avg_points` → Rolling mean of constructor strength
- One-hot encoding for Team
- `position_gain` calculated for EDA insight
  Cumulative features were shifted by 1 race to avoid leakage.

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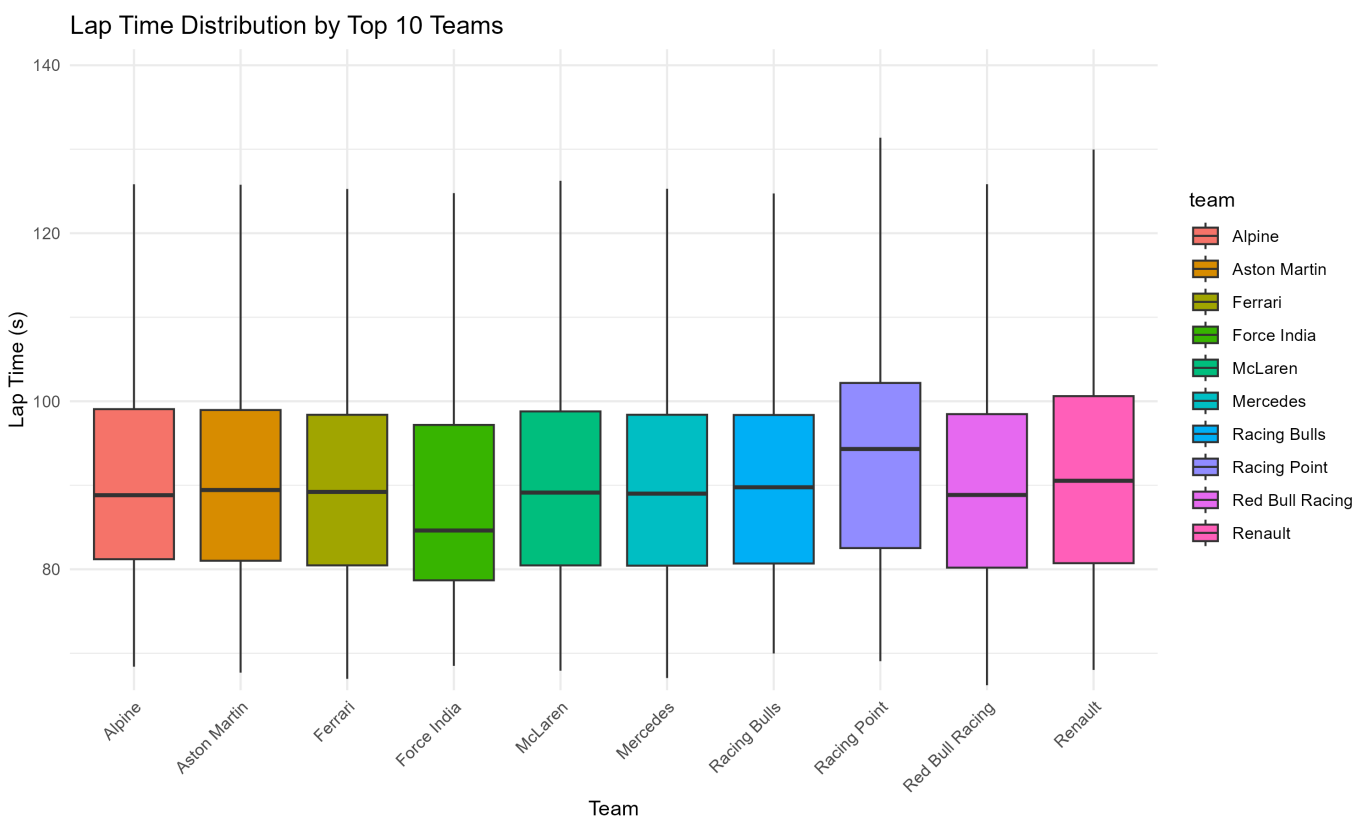
4. Exploratory Data Analysis (EDA)

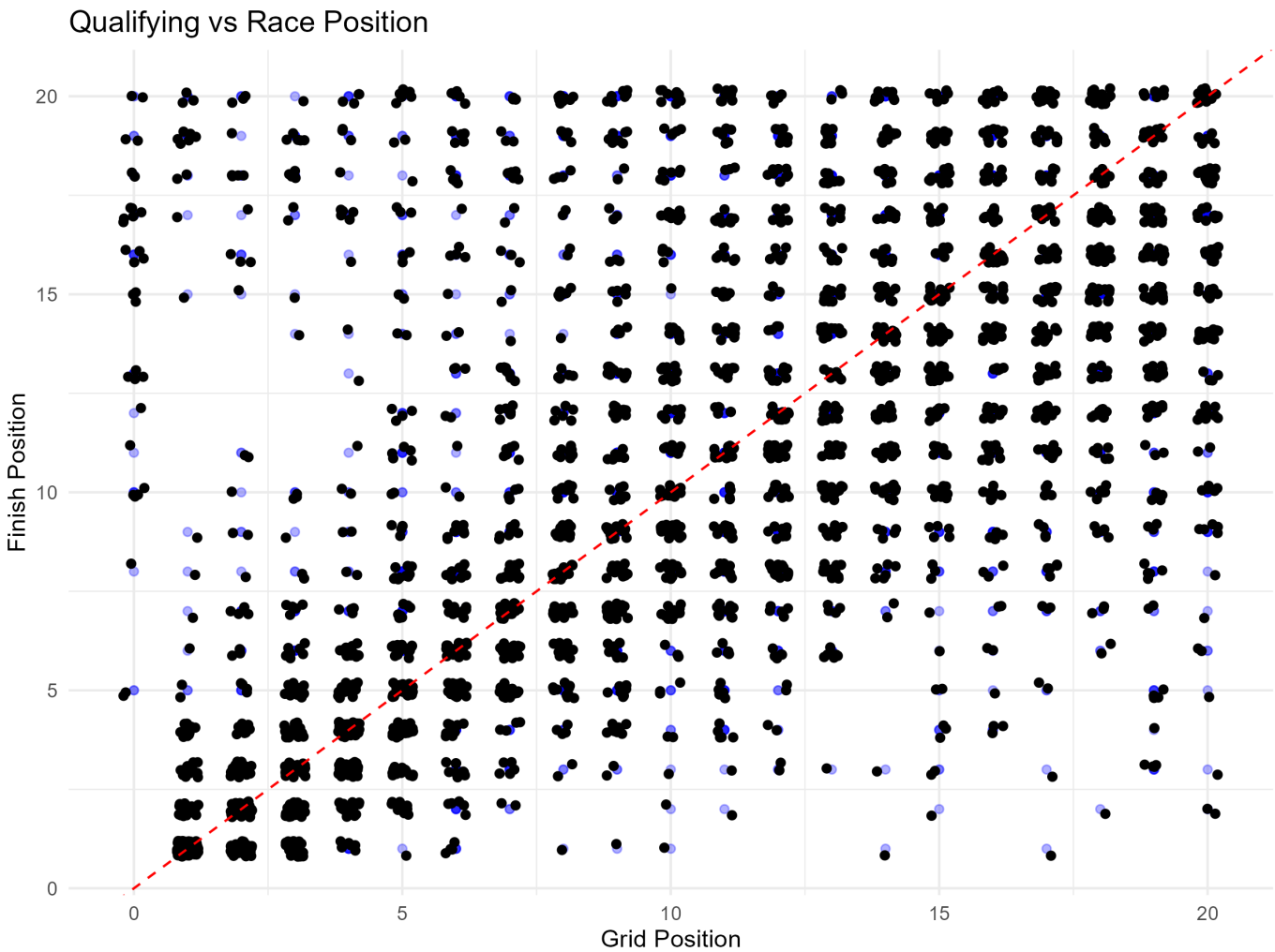
4.1 Lap Time Distribution by Team



This visualization shows performance consistency among top teams.
Observations:

- Top teams exhibit lower median lap times.
- Variance differs between midfield and front-running teams.
- Constructor strength significantly influences race pace.

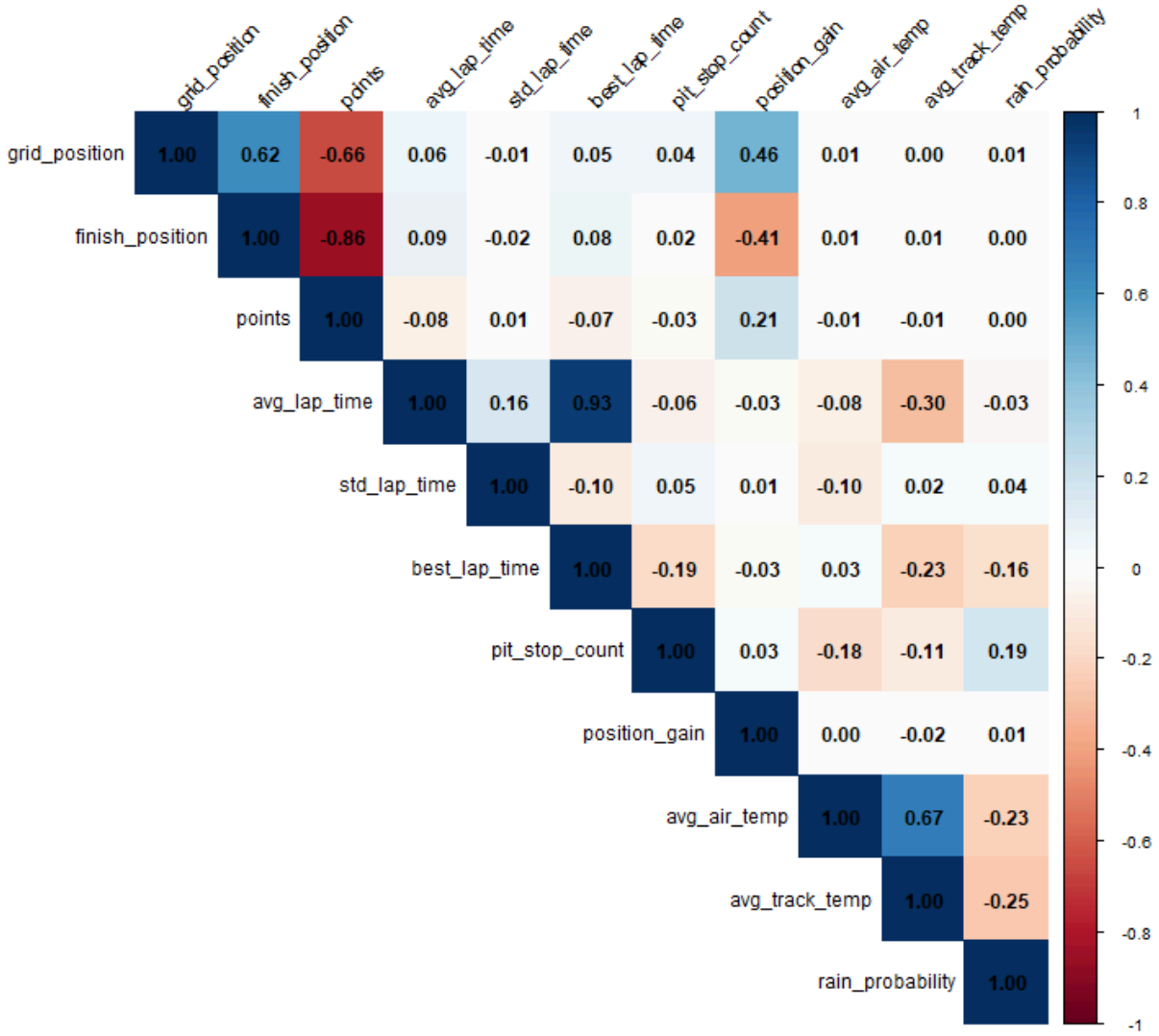
4.2 Qualifying vs Race Position



Strong positive correlation between grid and finish position is observed.
Insights:

- Track position is critical in modern F1.
- Deviation from the diagonal reflects strategy, DNFs, and incidents.
- Grid position is likely the strongest predictive feature.

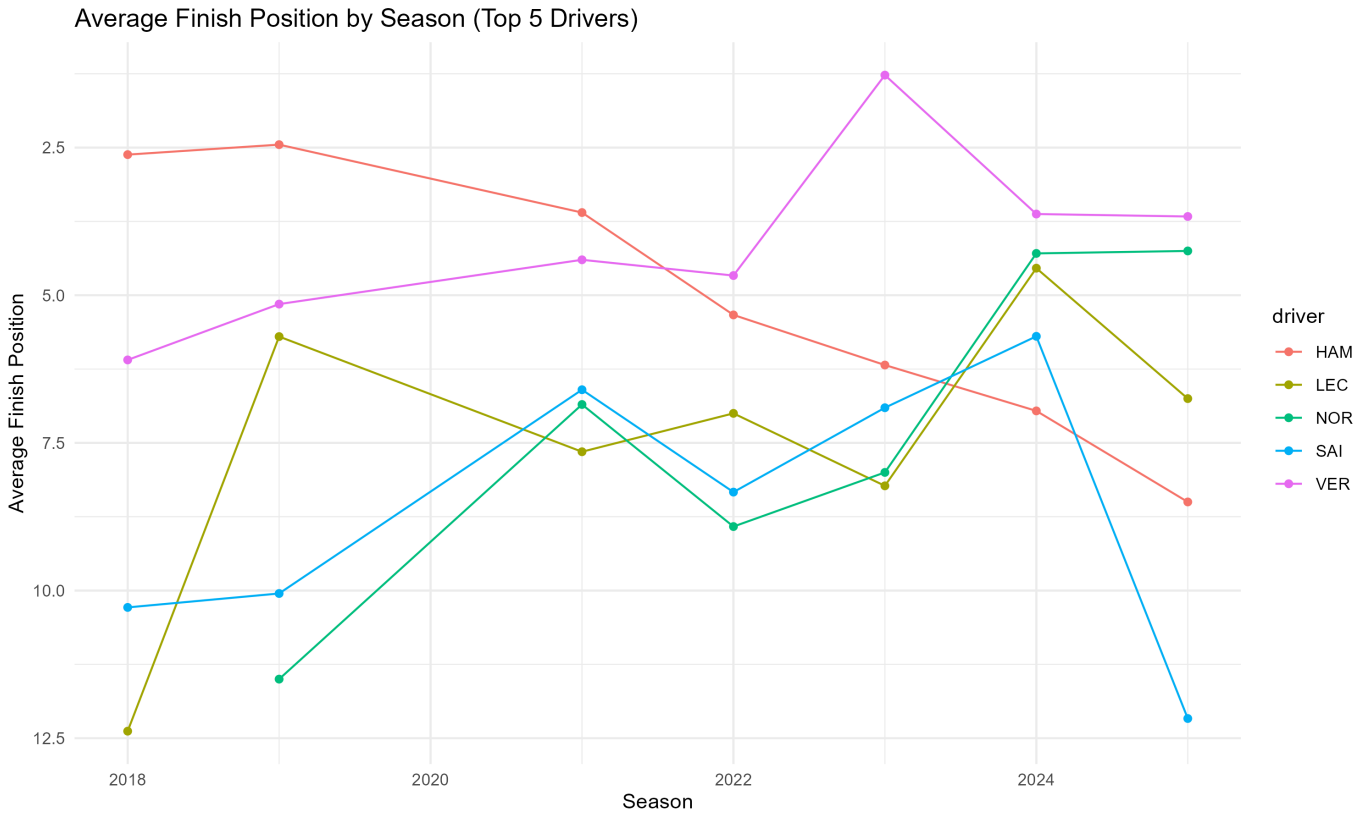
4.3 Correlation Heatmap



Key Observations:

- Strong negative correlation between finish_position and points.
- High multicollinearity between avg_lap_time and best_lap_time (~0.93).
- Moderate correlation between grid_position and finish_position.
- Weather variables show weaker correlation.
  This informed feature selection for modeling.

4.4 Driver Performance Trends Over Time



Insights:

- Driver performance trends align with regulation changes.
- Certain drivers show consistent dominance periods.
- Performance variability increases during regulation resets.

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5. Model Selection & Training

5.1 Algorithms Used

Two models were implemented:

1. **Linear Regression (Baseline)**
   - Assumes linear relationship.
   - Provides interpretability.
   - Serves as benchmark.
2. **Random Forest Regressor**
   - Captures non-linear relationships.
   - Robust to feature interactions.
   - Uses ensemble tree-based learning.

5.2 Training Strategy

Time-aware split:

- Training Set: 2018–2023

- Test Set: 2024–2025
  This prevents future information leakage.

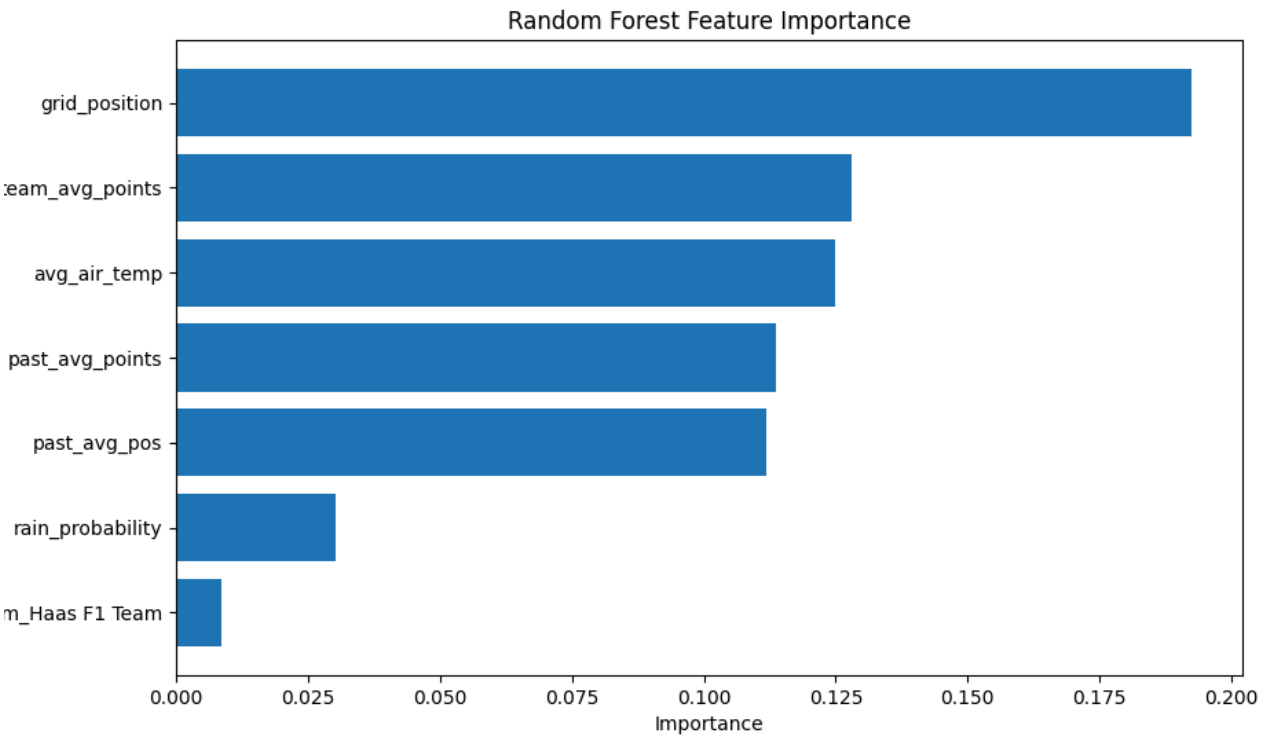
5.3 Model Performance (Test Set)

Model	RMSE	MAE	R^2
Linear Regression	4.35	3.49	0.43
Random Forest	4.60	3.68	0.36

Interpretation:

- Linear Regression outperformed Random Forest.
- Indicates a strong linear relationship between predictors (especially grid position) and finishing position.
- Random Forest may require hyperparameter tuning for improved generalization.

5.4 Feature Importance (Random Forest)



Top Predictors:

1. Grid Position (~0.19)
2. Team Average Points
3. Average Air Temperature
4. Past Driver Performance
   Grid position emerges as the dominant predictor of race outcome.

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6. Conclusion

This project successfully built an end-to-end machine learning pipeline to predict Formula 1 race finishing positions using data from 2018–2025.
Key Findings:

- **Qualifying position is the strongest predictor of race results.**
- Historical driver and team performance significantly improve prediction.
- Weather plays a secondary role.
- A simple linear model performs competitively due to strong linear dependencies.
  Model performance ($R^2 \approx 0.43$) demonstrates moderate predictive power, reflecting both structural determinants (grid, team strength) and inherent race unpredictability.

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7. Limitations & Future Work

Limitations

- Does not explicitly model DNFs or mechanical failures.
- No safety car or strategy simulation features.
- Hyperparameter tuning not extensively explored.
- Regulation-era segmentation not separately modeled.

Future Improvements

- Gradient Boosting or XGBoost implementation.
- Classification model to predict race winner probability.
- Separate modeling for dry vs wet races.
- Incorporating tyre strategy dynamics.
- Feature scaling and advanced cross-validation.